

Customer Segmentation using Machine Learning for Targeted Marketing

College: Dayananda Sagar College of Engineering

Name: Yashwanth H

USN: 1DS22ET111

1. Problem Statement:

In modern retail and e-commerce systems, businesses deal with large volumes of customer data but often fail to utilize it effectively for personalized marketing. Without proper segmentation, marketing strategies remain generic, leading to reduced customer engagement and lower profitability. The problem is to identify meaningful customer groups based on purchasing behaviour and demographic attributes to enable data-driven decision-making.

2. Dataset Description:

The project will use the *Mall Customers Dataset*, which contains structured customer information including:

- Customer ID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1–100)

This dataset is suitable for unsupervised learning as it does not contain predefined labels and allows pattern discovery.

3. Algorithm / Methodology:

The primary algorithm used will be **K-Means Clustering**, an unsupervised machine learning technique that partitions data into K distinct clusters based on feature similarity.

Steps involved:

- Data preprocessing and cleaning
- Feature selection (Annual Income and Spending Score)
- Feature scaling using StandardScaler
- Determining optimal number of clusters using Elbow Method
- Applying K-Means clustering
- Visualizing clusters using scatter plots
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4. Expected Output:

The model is expected to divide customers into distinct segments such as:

- High Income – High Spending (Premium Customers)
- High Income – Low Spending (Potential Customers)
- Low Income – High Spending (Impulse Buyers)
- Low Income – Low Spending (Low-value Customers)

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- Average Customers

The output will include:

- Clustered dataset with labels
- Visual representation of customer segments
- Insights for business strategy

5. Applications:

- Targeted marketing campaigns
- Customer retention strategies
- Personalized product recommendations
- Business decision support systems

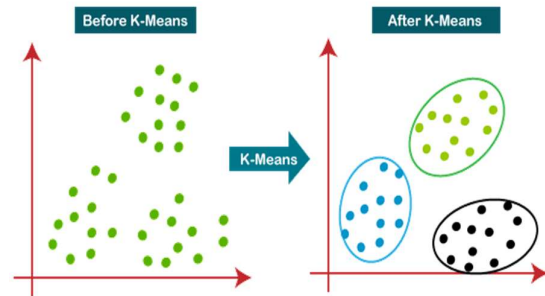


Fig.1 K-Means Clustering

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6. Conclusion:

This project demonstrates how K-Means clustering can effectively group customers based on their income and spending behaviour. The identified segments help businesses understand different types of customers and tailor their marketing strategies accordingly. It also shows the importance of preprocessing and selecting the right number of clusters for accurate results.

Furthermore, the model provides a simple yet powerful way to convert raw data into actionable insights. These insights can improve customer satisfaction, increase sales, and support better decision-making. The approach can also be extended with more features or advanced algorithms for deeper analysis in future work.